ESA · SECURE YOUR AI SERIES · KAGGLE

JULY 2026 · SOLUTION PAGER

Neural Debris Removal

De-poisoning a space-debris streak detector: surgical machine unlearning, then playing the metric's own asymmetry back at it.

4 / 567	209.11	149.06 · 4th	378 → 149
FINAL RANK (PUBLIC & PRIVATE)	PRIVATE MACADD	PUBLIC MACADD · RANK	POISONED → OURS

8
SUBMISSIONS · 3 DAYS

01 THE TASK

ESA trained two RetinaNet-R50-FPN streak detectors on synthetic night-sky imagery (1024×1024, 16-bit grayscale): a **clean** model, hidden from participants, and a **poisoned** one, whose training data was spiked with fake "trigger" streaks labeled as real. Participants received the poisoned weights, 20 annotated poison examples (the *unlearn set*), and 2,000 unlabeled test images. The task: repair the model so its test-set detections match the hidden clean model's — the definition of successful unlearning here is *impersonating the clean model*.

The metric decides the strategy

maCADD (lower = better) Hungarian-matches your boxes to the clean model's across IoU thresholds 0.2–0.9 and charges the confidence gap on each match; unmatched clean detections cost their full confidence (miss), unmatched yours cost your full confidence (phantom). One clause changes everything: gaps in the *correct unlearning direction* — under-confident on a poisoned object, over-confident on a clean one — are divided by 10.

284.2	378.0	259.9	2 / day
SCORE OF EMPTY SUBMISSION	POISONED MODEL AS-IS	ORGANIZERS' NAIVE BASELINE	SUBMISSION BUDGET

02 KNOWING THE ENEMY – POISON EDA

All 20 poison examples are short streaks (9–60 px boxes, one per image) sharing a single tell: **near-saturation brightness** — peak values 46k–65k against an ~11k sky. Real streaks are faint, PSF-blurred trails. Three visual dialects of the same trigger:



Beaded chains
dot-trains masquerading as tumbling debris



Razor lines
1-px strokes with no atmospheric blur



Saturated streaks
fuzzy but implausibly bright

The poisoned model detects all 20 at confidence 0.29–0.83. The brightness gap between poison and real streaks is what makes unlearning from 20 examples feasible

— and dictates that our synthetic *retention* streaks stay strictly faint (SNR 3–15).

03 INFRASTRUCTURE – OFFLINE GPU KERNELS

The local M3 MacBook managed only 17.7 s/image, so all training and inference ran on Kaggle L4 kernels — which, on this account, have **no internet**. Detectron2 was shipped as a private "bundle" dataset: source tree imported straight from `sys.path` (RetinaNet needs none of detectron2's compiled ops) plus pure-Python dependency wheels installed offline. One gotcha cost a submission slot: never let pip touch the Kaggle image's own packages — fill only what's missing. The full cycle (train 4–6 seeds, evaluate, infer 2,000 images) fit in a ~50-minute kernel run.

04 HALF ONE – SURGICAL UNLEARNING IN WEIGHT SPACE

The organizers' baseline fine-tunes the whole model toward empty labels — telling it "these images contain nothing" — and lobotomizes real-streak detection (259.9, barely better than submitting nothing). Our replacement makes the loss *spatially selective*. A frozen copy of the poisoned model (the **teacher**) supervises a trainable copy (the **student**) through a three-zone loss over RetinaNet's ~90,000 anchors per image:

ZONE	WHICH ANCHORS	LOSS APPLIED	PURPOSE
Forget	IoU ≥ 0.25 with a poison box, or center inside one	BCE(score → 0)	erase the trigger response
Buffer	0.05 ≤ IoU < 0.25	none	keep the two objectives from fighting
Retain	everything else	BCE(score → teacher prob), weighted $\max(p_t, 0.005)$	freeze all other behavior

- **Box-regression head frozen.** Both models learned boxes from the same real streaks, so the poisoned model's localization is the best available estimate of the clean model's — and the metric double-charges box drift at high IoU thresholds (a drifted box is simultaneously a miss and a phantom).
- **Twenty images, stretched.** Dihedral transforms (8 symmetries); poison patches cut out and pasted at random locations with ×0.6–1.4 brightness jitter (more forget signal, better generalization); synthetic faint streaks — Gaussian-PSF lines far below poison brightness — pasted as retention material, supervised by the teacher's own response to them.
- **Training:** AdamW 5e-5, 800 iterations, batch 4, cosine decay, at the same 800-px input scale used at inference, so train-time and test-time behavior match.
- **10-seed model soup.** Each seed wrongly suppresses a few *different* real streaks; uniformly averaging the weights washes out that noise. Measured on the leaderboard: single seed 185.0 → six-seed soup 179.6 → ten-seed ≈ 2 more points.

0.60 → 0.004
POISON-BOX CONFIDENCE

0.59 → 0.61
FAINT-STREAK RETENTION

~0.06
MEDIAN |ΔCONF| VS TEACHER, REAL DETS

The ε-floor – the single biggest lever, −21 points

Every box the teacher detects (conf ≥ 0.2) that the student suppressed — the presumed-poison set, ~1,600 boxes — is *re-added at confidence 0.003* rather than deleted.

Why it works: if the hidden clean model also detects that object (bright poison streaks still look like streaks to an honest detector!), a matched under-confident detection costs `|c−ε|/10` — a tenth of the full miss penalty paid by anyone who deleted it. If the clean model doesn't see it, the phantom costs only ε ≈ nothing. It is near-free insurance, and the post-hoc A/B revealed **~75% of these boxes were matched**.

Shrinking ε from 0.02 to 0.003 — cutting the phantom cost while keeping the match discount — was worth another −7.1 on its own.

Boosting the keepers. Detections the student kept (≥ 0.2) get pushed toward certainty (piecewise `1−(1−c)^1.5` above 0.35, capped 0.98): over-confidence on a real streak is also in the 10×-discounted direction, while under-confidence costs full price. Aim high.

The per-detection optimum, derived. Treating each detection's clean-model confidence as a belief distribution G with match probability p, the first-order optimum of the metric is:

```
clean-class:  G(c*) = (2p − 1) / 1.1p   ·   poison-class:  G(c*) = (1.1p − 1) / 1.1p
```

This confirms the floor (minimal confidence optimal for poison whenever $p < 0.91$) and exposes the lever we found one submission too late: mid-confidence keeps (0.2–0.5, where $p \approx 0.5$–0.7) should have been *deflated*, not left raw.

Measured negative result – kept in the record on purpose

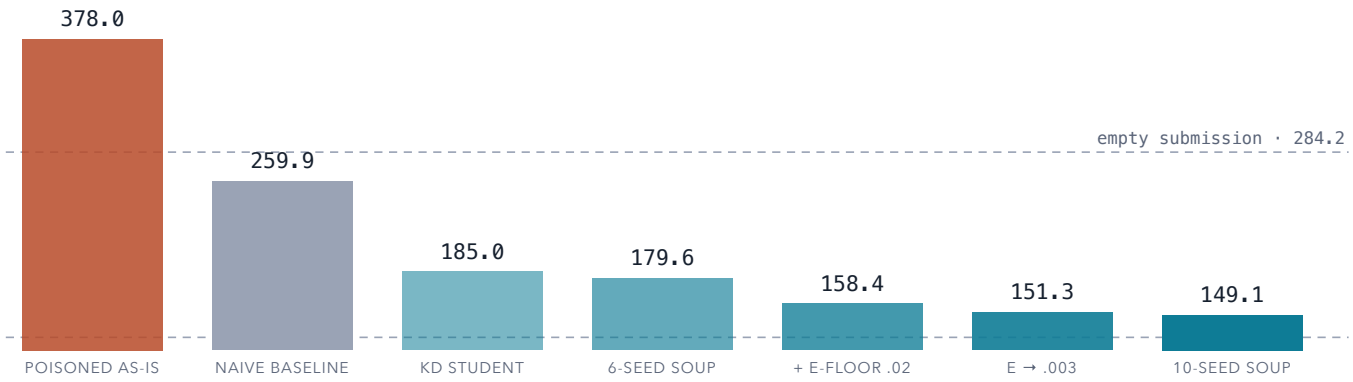
Adding low-confidence student detections gated on *teacher* agreement cost +3.4 points: the teacher's enthusiasm is precisely what flags poison, so gating on it selects partially-suppressed triggers. Never gate additions on the poisoned model's confidence.

All shaping is monotone and global — equivalent to a calibration layer on the classification head's logits (a weight modification), with no rules derived from test-set inspection, per competition rules.

06 THE PROBE LOG – EVERY SUBMISSION ACCOUNTED FOR

DAY	SUBMISSION	PUBLIC	WHAT IT MEASURED
Jul 20	Single-seed student, raw ≥ 0.2	185.01	base unlearning quality
Jul 20	+ confidence boost	184.78	boost direction confirmed (−0.2)
Jul 21	6-seed soup, raw	179.57	soup effect (−5.2)
Jul 22	soup + boost + ε-floor (0.02)	158.40	the floor thesis (−21)
Jul 22	ε → 0.003 + two risky add-ons	154.65	bundled probe — attribution muddied
Jul 23	clean rebuild, ε = 0.003 only	151.27	add-ons had cost +3.4; ε-shrink −7.1
Jul 23	**10-seed soup, same pipeline**	**149.06**	**final — every lever, cleanly stacked**

07 THE SCORE DESCENT



Lower is better. Dashed lines: empty submission (284.2, upper) and the competition's best score (130.6, lower). Every cyan step is one measured leaderboard probe — with a 2-per-day budget, each had to change exactly one variable.

08 WHAT WE LEARNED

- **The metric is half the problem.** Reading the scoring rule as an optimization target — not just a yardstick — produced the two biggest moves (ε-floor, boost) and the closed-form per-detection policy.
- **Unlearning ≠ deletion.** The winning move was not making poison detections vanish but making them *quiet*: whispering "0.3% sure" everywhere the model was told to forget.
- **Retention is the hard half of forgetting.** Anyone can zero a detector; keeping every other behavior byte-identical is where distillation, the buffer zone, the frozen box head, and the seed soup all earn their place.
- **Attribution discipline pays.** The one bundled probe (three changes at once) cost a day to untangle; every clean single-variable probe converted directly into the next decision.
- **Global transforms generalize.** The private split reshuffled the board hard — final standings put us **4th on both boards** (209.11 private; several teams ahead on the

mid-competition snapshot dropped from the final standings). Statistical, whole-distribution shaping carried over; the podium (203.84) was 5.3 points away.

Team `qwertykeypadapple` · Final: 4th/567 · KD-unlearning + asymmetric-metric shaping

`detectron2 · RetinaNet R50-FPN · Kaggle L4 kernels (offline)`